

The Margin-Distortion Project, explained.

A plain-language walkthrough of the **finance theory** and the **datasets** behind Project 5 — and a simple, jargon-free version of how the software turns a pile of spreadsheets into a shareable database. No code required to follow along.

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03 Theory · why R&D hides profit

04 Theory · how companies fund themselves

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07 The tech, dumbed all the way down

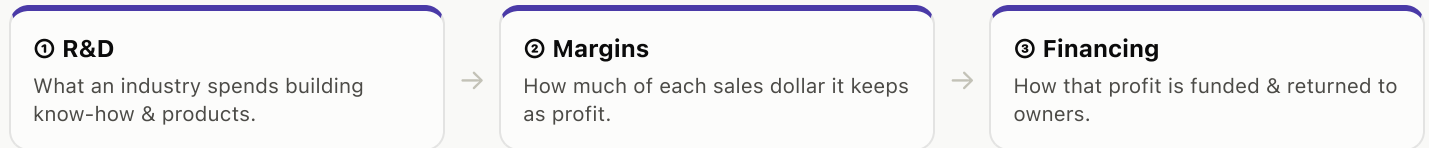
08 What you can ask it · how to view

Data © Aswath Damodaran (NYU Stern), free for educational use · Jan-2026 vintage (trailing 12 months ≈ Q3 2025) · github.com/ahmed3142/fin

01 The big idea, in one breath

What the project actually asks — and why it's interesting.

Every company does three things in a chain: it **invests** (research & development), which — if it works — produces **profit** (margins), which then gets **funded and handed back** to shareholders (dividends, buybacks, debt). Professor Aswath Damodaran publishes free data on all three, broken down by industry. Nobody usually looks at them *together*.



This project joins those three tables (plus valuation tables) into one dataset and asks a single, pointed question:

THE THESIS

Reported profit margins make R&D-heavy industries look worse than they really are — so ranking industries by reported margins is misleading. Does the stock market already know this?

The short answer we found: **yes, mostly**. The market's price of a company (its EV/Sales ratio) lines up a little better with the *R&D-adjusted* margin than the reported one (**R^2 0.425 vs 0.406**). The distortion is real but **concentrated in a handful of R&D-heavy industries** — software, semiconductors, pharma — not spread across all of them.

WHAT THIS IS

A careful, reproducible comparison of ~94 U.S. industries, every number traceable to a primary source, with the limitations stated up front.

WHAT THIS IS NOT

A stock tip, a firm-by-firm claim, or proof of cause-and-effect. These are industry averages — a description of the landscape, not advice.

Analogy

Think of each industry as a **restaurant**. "Reported margin" is the profit after counting this year's kitchen remodel as a pure loss. "Adjusted margin" spreads that remodel over the years it will actually help. The remodel-heavy restaurants (R&D-heavy industries) look unfairly unprofitable on the first measure — and we test whether diners (the market) already price them fairly.

02 The theory that ties it together

Why R&D, margins, and financing are really three views of one thing: a company's age.

Damodaran's signature idea is the **corporate life cycle**. A company (or a whole industry) moves through stages, and at each stage its investment, profit, and funding behave in a predictable, linked way. So an industry's position in one of our three tables should match its position in the others — they're three clocks telling the same time.

Young / Growth	Mature Growth	Mature Stable	Decline
R&D: very high — the whole point	R&D: still high, moderating	R&D: lower	R&D: shrinking
Margins: thin or negative	Margins: rising, moats forming	Margins: fat & stable	Margins: compressing
Funding: issues stock (no cash yet)	Funding: starts self-funding	Funding: buybacks + dividends; adds debt	Funding: repays debt, pays out cash

■ Why young firms *issue* stock and mature firms *buy it back*

This follows the **pecking-order theory**: companies prefer to fund themselves with internal cash first, then borrowing, and only issue new stock as a last resort (issuing shares signals the stock might be overpriced, and it dilutes existing owners). Young R&D companies have *no* internal cash and little borrowing capacity — so they're forced into that last-resort corner and issue equity. Mature companies have plenty of cash, so they do the reverse: buy their own shares back.

Analogy

A student (young firm) borrows and takes on housemates (issues equity) because they have no savings. A established earner (mature firm) pays down the loan and buys the housemates out (buybacks + debt repayment). Same person, different life stage — that's an industry moving down the table.

WHY IT MATTERS FOR THE PROJECT

Because the stages are linked, the *interesting* industries are the ones that break the pattern — e.g. a mature, cash-rich industry that **still** pours money into R&D, or one that funds its buybacks by **borrowing** rather than from profit. Joining the tables is what lets us spot those.

03 The accounting trick at the heart of it

Why reported margins understate R&D-heavy industries — and how Damodaran corrects it.

When a company builds a **factory**, accounting rules spread its cost over the many years it will be used (this is called *capitalizing* and *depreciating*). But when a company spends on **R&D**, the rules force it to count the *entire* cost as an expense in the year it's spent — even though that research builds value for years to come.

The result: a company spending heavily on R&D looks **less profitable than it economically is**, because money that is really an *investment* is being subtracted from this year's profit.

■ The two ways to count R&D (illustrative)



Reported subtracts *all* of this year's R&D now. Adjusted treats R&D like the factory — spread over its useful life — so the margin reflects the real economics. The gap between the two bars **is the distortion**, and it is widest exactly where R&D spending is highest.

Damodaran does this correction for us: he estimates an R&D "asset," spreads it over an industry-specific life (~5 years for software, ~10 for pharma), and republishes an **adjusted operating margin** right next to the reported one. Our project measures the gap between them across every industry.

■ The test — and the honest result

If reported margins mislead, then the market — which cares about real economics — should price companies in line with the *adjusted* margin, not the reported one. We checked by comparing each industry's market valuation (EV/Sales, P/E) against both margins:

- **EV/Sales fits the adjusted margin slightly better** (R^2 0.425 vs 0.406) — mild evidence the market sees through the accounting.
- The adjustment **reranks only 1 of 84 industries** by 5+ places (rank correlation 0.998) — so the distortion doesn't reshuffle the whole league table...
- ...it's **concentrated in the high-R&D tail** (software, semis, pharma). That's the honest, non-hyped finding.

THE GUARDRAIL THAT KEEPS IT HONEST

The adjusted margin is *built from* Damodaran's R&D estimate — so "proving" R&D drives the adjusted margin would just be re-deriving his own arithmetic (circular). We never do that; we point the test at an **independent outcome** — the market price. The code even has an automatic check that **refuses** to run the circular comparison.

04 Dividends, buybacks & the funding signature

The third link in the chain — and two subtle traps the data reveals.

Once an industry is profitable, it faces a choice: keep the cash, hand it out as **dividends**, or **buy back its own shares**. The mix is a fingerprint of maturity.

■ Dividends

Sticky and rarely cut — so paying them signals management believes profits are **permanently** high. Common in old, stable industries (utilities, staples).

■ Buybacks

Flexible and tax-friendly — used to return *spare* cash and signal the shares are cheap. Dominant in tech & pharma.

The data file tracks all of it: dividends, buybacks, new shares issued, and debt raised or repaid. One especially useful number is **Net Equity Change = shares issued – shares bought back**. Negative means the industry is a net *buyer* of its own stock (returning capital); positive means it's a net *issuer* (raising it).

■ Two traps hiding in the numbers

TRAP 1 · LEVERAGE-FUNDED BUYBACKS

Some mature industries buy back huge amounts of stock **while simultaneously borrowing more**. That's capital return funded by *debt*, not by profit — a very different (and riskier) story than it looks at first glance. Joining the financing table to the debt columns exposes it.

TRAP 2 · BUYBACKS THAT AREN'T REALLY "RETURNS"

Industries that pay staff heavily in stock (high stock-based compensation) issue lots of new shares to employees, then buy shares back just to *offset* that dilution. Their gross buybacks look generous but much of it is treading water — so **gross buybacks overstate the cash actually returned**. The "Net Equity Change" number strips this out.

THE CONNECTING THREAD

Notice how all three theory sections rhyme: **young → invest & issue equity; mature → fat margins & buy back stock; declining → shrink & repay debt**. The whole project is really one question — *where is each industry on that arc, and does the data agree with itself?*

05 The datasets: what they are

Free, industry-level, and all keyed to the same 94 industries.

Aswath Damodaran is an NYU Stern finance professor who publishes ~30 free datasets every January, each summarizing U.S. public companies **by industry**. Crucially, every file uses the **same ~94-industry list** (alphabetically Advertising → Utility (Water)) — that shared list is what lets us snap them together like Lego. The numbers are trailing-12-months through about Q3 of the prior year, so the "2026" files really describe **late 2025**.

■ The three core files (the value chain)

FILE	MEASURES	KEY COLUMNS WE USE
margin.xls	Profitability	Gross & net margin, reported operating margin, Damodaran's R&D-adjusted operating margin, R&D/Sales, stock-comp/Sales
R&D.xls	Investment	R&D as % of revenue, capitalized-R&D estimate, R&D 1–5 years ago, 5-yr R&D growth
finflows.xls	Funding	Dividends, buybacks, equity issued, net equity change, debt raised/repaid, lease debt

■ The valuation files (the market's verdict)

FILE	MEASURES	KEY COLUMNS WE USE
psdata.xls	Price vs sales	Price/Sales, EV/Sales (the market-price test)
pedata.xls	Price vs earnings	Current / trailing / forward P/E, PEG, expected growth
divfcfe.xls	Payout capacity	Payout ratio, free cash flow to equity, cash returned vs FCFE

ONE NUMBER TO KNOW: "VALUE-WEIGHTED"

Each industry figure is a **big-company-weighted average** (add up all firms' dollars, then divide) — *not* the typical firm. So "Software margin" is effectively Microsoft + Oracle + Adobe, not a mid-sized studio. It describes the **industry as a portfolio**, which is exactly what we want for a top-down comparison — but it's why we never claim anything about an individual company.

06 The gotchas we caught

Free data is messy. Finding and fixing these *is* the craftsmanship — and each one is real.

1 · The download links point to the wrong files

The web page for the R&D data links to `roe.xls`; the financing page links to `fundgrEB.xls`. Trusting the buttons gets you the wrong dataset entirely.

✓ **Fix:** go straight to the real files at `/pc/datasets/<name>.xls`.

2 · A real spelling mistake in the data

One industry is spelled `Heathcare Information and Technology` (missing the "l") in some files but correctly in others. Match the files on that raw text and the industry *silently vanishes* — no error, just a wrong answer.

✓ **Fix:** give every industry a stable ID via a spell-correcting lookup, and **halt loudly** if any name doesn't match.

3 · A definition that contradicts the data

The financing file's own notes say "Net Equity Change = issuance – (dividends + buybacks)". The actual numbers say **issuance – buybacks** (dividends not subtracted). We checked the arithmetic row-by-row and trusted the data, not the note.

✓ **Fix:** document the true formula; compute real net payout separately when needed.

4 · Hidden "Total Market" summary rows

Each file ends with 2 grand-total rows. Leave them in and they act like a giant 95th "industry" that swamps every average and chart.

✓ **Fix:** strip any row containing "Total Market" before doing any statistic.

5 · Percentages vs. multiples

Margins are stored as decimals ($0.23 = 23\%$), but EV/Sales and P/E are *multiples* ($6\times$, $20\times$) — not percentages. Rescale them the same way and a " $6\times$ EV/Sales" becomes a nonsensical "0.06".

✓ **Fix:** tag each column's type so multiples are left exactly as published.

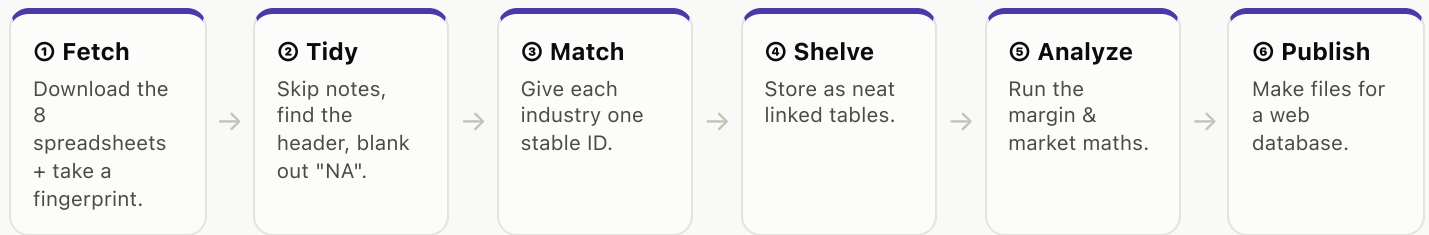
6 · The files are overwritten every January

Damodaran replaces each file in place each year at the same web address — so re-downloading silently changes "last year's" numbers, and your past results become impossible to reproduce.

✓ **Fix:** save a dated copy and a digital fingerprint (hash) of every file we used.

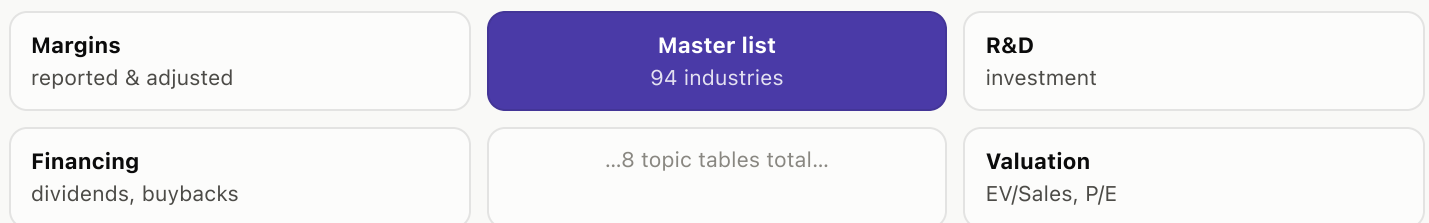
07 How the software works (no jargon)

The whole program is an assembly line that turns messy spreadsheets into one clean, shareable database.



■ "Shelving" the data — the one idea worth knowing

Instead of one giant messy spreadsheet, the data is stored as a **master list** of industries plus one small **topic table** for each subject, all pointing back to that master list. This layout is called a *star schema* — picture the master list in the middle and the topic tables around it like points of a star.



Analogy

It's a **library**. There's one master catalogue (the list of industries), and separate subject shelves (margins, R&D, financing...) that all reference the same catalogue number. Ask "everything about Semiconductors" and the system instantly gathers one card from each shelf.

■ The plain-English glossary

DuckDB	A lightweight database that lives in a single file on the laptop — a spreadsheet that can do heavy lookups instantly. Used to <i>build</i> everything.
Postgres	The standard "serious" database format companies use. We export the finished data into it.
Supabase	A free website that hosts a Postgres database and gives you a click-through table viewer — so anyone can browse or query the data in a browser.
Quality gates	Automatic inspectors that refuse to build unless everything's right: exactly 94 industries, every name matched, no "Total" rows, firm counts add up, and the circular analysis is blocked.

WHY THIS EARNS TRUST

One command rebuilds the entire thing from scratch, **15 automatic tests** confirm the maths and the data checks, and every source file is fingerprinted — so the results are **reproducible**, not a one-off spreadsheet nobody can re-run.

08 What you can ask it, and where to look

The finished database answers real questions in one line each.

■ Questions the data can answer

- **Which industries are most distorted by R&D accounting?** — sort by the gap between reported and adjusted margin (the R&D-heavy tail rises to the top).
- **Does the market price the reported or adjusted margin?** — compare each against EV/Sales and P/E.
- **Who funds buybacks with debt?** — industries buying back stock *and* borrowing at the same time.
- **Which industries look like "growth" vs "mature"?** — high R&D + equity issuance vs fat margins + buybacks.

THE HEADLINE, ONE MORE TIME

Across 84 non-financial industries, the market's EV/Sales tracks the **R&D-adjusted** margin a little better than the reported one (**R^2 0.425 vs 0.406**). The margin adjustment reshuffles almost nobody (rank correlation 0.998) — so the distortion is **concentrated in the high-R&D tail**, and that's exactly where screening on reported margins goes wrong.

■ Where everything lives

YOU WANT...	LOOK AT...
This explainer & the deck	docs/pdf/
The full written brief	docs/briefs/05-...md
The pipeline code	projects/05-industry-value-chain/src/
Ready-to-load database files	.../sql/postgres/
Step-by-step Supabase guide	.../docs/SUPABASE.md

SEE IT IN A BROWSER

Open a free Supabase project → SQL Editor → paste the four `.sql` files in order → the whole dataset appears as clickable tables, with an automatic web API. Full instructions in `SUPABASE.md`. Everything is free, and no private keys ever leave your machine.